C29xMCAL_01.03.00 Release Notes

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Introduction

F29x MCAL is a cohesive set of tools for AUTOSAR based development on C2000 real-time controllers. It includes device-specific drivers (MCALs), EB Tresos Plugins for configuration, Bare-metal Examples for each driver.

F29H85x MCAL v01.03.00 is an early adopter release targeted for F29H85x.

Device Support

The following devices are supported with this release

- MCU's
  - F29H859TU-Q1

- Hardware Platforms
  - F29H85X-SOM-EVM

What Is Supported

- MCAL / CDD (AUTOSAR **v4.3.1**):

Drivers	**Modules**

Micro controller Drivers	GPT, MCU, WDG
Communication Driver	CAN, LIN, SPI
I/O Drivers	DIO, PORT
Memory Drivers	FLS
Complex Device Drivers	ADC, XBAR, SENT, IPC, UART, PWM, ECAP

- Reference Code:
  - Multi-Core startup code
- Build System:
  - CMake
  - CCS Projectspec

Note: The minimum version of CCS required is CCSv20.02.00 and compiler version is C29Clang 2.0.0 STS. Please refer to the Dependencies section for the download links.

What Is Not Supported

- Multi-core MCALs as per AUTOSAR 4.3.1.

New In This Release

- MCAL / CDD :
  - Existing modules with bug fixes -
    - PORT
    - FLS
    - DIO
    - CDD IPC
    - CDD XBAR
    - CDD UART
    - SPI
    - MCU
    - GPT
    - LIN
    - CDD ADC
    - WDG

- Existing modules with improved design -
  - CAN
- New Functionality added
  - CDD PWM
  - CDD ECAP
- Emulation support for F29P58x
- Emulation support for F29P32x

Module Version Number

MCAL / CDD	Module Version Number
MCU	02.00.00
PORT	02.00.00
DIO	01.00.03
GPT	02.00.00
CAN	02.00.00
WDG	02.00.00
LIN	02.00.00
SPI	02.00.00
FLS	02.00.00
CDD ADC	02.00.01
CDD XBAR	02.00.02
CDD SENT	02.00.01
CDD PWM	02.00.00
CDD ECAP	02.00.00
CDD IPC	01.00.02
CDD UART	02.00.00

Note:

We follow Autosar's requirement [SRS_BSW_00321] for versioning of each module-

5.2.4.10 [SRS_BSW_00321] The version numbers of AUTOSAR Basic Software Modules shall be enumerated according specific rules

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Type:	Valid
Description:	The version numbers of AUTOSAR Basic Software Modules shall be enumerated according to the following rules: - Increasing a more significant digit of a version number resets all less significant digits - The PATCH_VERSION is incremented if the module is still upwards and downwards compatible (e.g. bug fixed) - The MINOR_VERSION is incremented if the module is still downwards compatible (e.g. validfunctionality added) - The MAJOR_VERSION is incremented if the module is not compatible any more (e.g. existing API valid)]
Rationale:	Provide unambiguous version identification for each module, provide version cross check as well as basic version retrieval facilities. Compatibility is always visible!
Use Case:	Example: ADC module with version 1.14.2:

Fixed In This Release

Key	Summary	Component/s	Severity	Status
MCAL-32230	LIN: Wakeup interrupt is calling EcuM_CheckWakeup with wrong argument	Lin	S3-Minor	Closed
MCAL-32153	CAN: More than 256 Hoh can't be handled in driver	Can	S3-Minor	Closed
MCAL-32095	CAN: CanIf_TriggerTransmit is called with wrong PduId	Can	S3-Minor	Closed
MCAL-32094	CAN: RX FIFO handling is incorrect	Can	S3-Minor	Closed
MCAL-32093	CAN: Incorrect filter configuration	Can	S3-Minor	Closed
MCAL-31557	TPS_ECUC_08011: Defining symbols used by EcuM/BswM to initialize BSW modules	Can	S3-Minor	Closed
MCAL-31556	Dependency to inclusions in Os.h file	Can, CDD ADC, CDD PWM	S3-Minor	Closed
MCAL-31540	CAN: Periodic read/write based on mapped function is not respected	Can	S3-Minor	Closed
MCAL-31536	Completion Notification to comm stack should be done at the end of ISR	Can	S2-Major	Closed

MCAL-31513	LIN: Not able to transmit master response frame after slave response frame	Lin	S2-Major	Closed
MCAL-31479	Multiple Watchdog instance support in EB Tresos	Wdg	S3-Minor	Closed
MCAL-31465	SpiMaxSequence textboxes are not auto filled after click AutoCal	Spi	S2-Major	Closed
MCAL-31463	CAN frame missing	Can	S2-Major	Closed
MCAL-31462	CAN RX FIFO overflow issue.	Can	S3-Minor	Closed
MCAL-30391	Fls_App_Example doesn't run in RAM mode	Fls	S2-Major	Closed
MCAL-30035	Mixed mode handling is incorrect in Can	Can	S3-Minor	Closed

16 issues

Known Issues And Limitations

There are few limitations and known issues associated with this release

Limitations

- CDD ECAP -
  - In case of high frequency signals (i.e. 500Khz and above), software overhead in user callback shall be kept to minimum to avoid missing edges.
- Test Infrastructure : Following tests have not been performed due to limitations of available test infrastructure -
  - CDD SENT : Following modes in SENT are not tested -
    - Enhanced 16-bit Slow Mode.
    - Nibble Count 1,2,3,4,5.
    - MTP Mode with all 4 sensors connected.
  - CDD IPC : IPC is validated between CPU1 and CPU3. Since, CPU2 is configured in lockstep in our framework; IPC between CPU1 and CPU2 or CPU2 and CPU3 is not validated.
  - Fls : Timeout tests are not performed as FLASH characterization data is not yet available for this device.

  - Spi : Board to board Spi communication validated upto 40MHz. 50MHz have been only test in loopback mode.
- Compiler
  - Link Time optimization is currently not supported.

New Known Issues

Key	Summary	Component/s	Workaround	Severity
MCAL-32322	CAN: Can communication with 5Mbps data rate is not verified	Can	Baudrate can be configured upto 4Mbps.	S3-Minor
MCAL-32321	All Modules: Link time optimization is not verified for all modules	All	By default LTO is disabled, user shall not enable it.	S2-Major
MCAL-31535	Not able to send LIN frames after calling Lin_Wakeup()	Lin	This is an IP limitation; the IP is unable to transmit any LIN frames after sending a wakeup signal. As a temporary workaround, customer can call Lin_WakeupInternal(), which doesn't transmit any wakeup signals on the Lin bus instead of Lin_Wakeup().	S2-Major
MCAL-31442	Incorrect uniqueness check for User Callback function in CDD SENT	CDD SENT	Configure dummy function uniquely.	S3-Minor

4 issues

Existing Known Issues

Key	Summary	Component/s	Workaround	Severity

No issues found

Dependencies

This release has dependency on the following tools

- CCS v20.02.00
- C29Clang Compiler 2.0.0 STS.
- EB Tresos (ACG 8.8.12) : Request access here.

References

Technical Support

https://e2e.ti.com/support/microcontrollers/c2000/f/171